SIGNIFICANCE

Earlier Intervention Necessitates Earlier Identification

Autism spectrum disorder (ASD) is a lifelong condition that affects ~2.3% of the population and carries a lifelong financial cost of $1.4-2.4 million per individual, thus representing a pressing public health challenge[29,30]. Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the individual, enhancing the quality of life for the family, and reducing a lifelong financial cost[31,32]. Unfortunately, with ASD diagnosis and therefore intervention onset occurring after age 4-6 years[33,34], we are missing out on critical periods of early intervention that could capitalize on peak neuroplasticity within the first 2 years of life[35–37]. **This is a critical barrier in the field**–earlier identification of ASD within the first year of life can *catalyze the onset of very early intervention,* leading to greater *improvements in long-term outcomes*. Addressing this critical barrier by identifying the earliest point of developmental divergence in ASD is a high priority.

The search for biological and behavioral markers of ASD has resulted in increased understanding of altered neurodevelopmental pathways, and **atypical attention** is among the most promising biomarkers that persists across the lifespan[38–42]. Research on infant attention suggests that infants with ASD exhibit reduced neonatal attention[6,43], diminished social monitoring[44,45], disrupted engagement of social sustained attention[46], and a developmental decline in eye-looking[7], all prior to diagnosis. **Still, this evidence is limited to <u>behavioral</u> studies of attention, and few extend to earlier than 2-3 months of age.** To advance our understanding of disrupted attention in ASD, there is a need to examine specific biological substrates associated with this developmentally persistent phenotypic feature. The <u>timing</u> and <u>context-specificity</u> of attention differences measured at <u>multiple levels</u> (e.g., behavioral, physiological) in <u>early development</u> may reveal key neurodevelopmental mechanisms in ASD. This project takes the critical next step in examining **biological underpinnings of disrupted attention in ASD** by testing whether reduced autonomic regulation of attention ("heart defined attention") is **a key early-emerging feature associated with ASD. Significantly,** this work will:

1. **Advance current etiological theories of ASD** that link, with heightened specificity, an established, lifelong cardinal feature of ASD – atypical attention – to autonomic regulation of attention in the first months of life.
2. **Pave the way for new developmental models of ASD** wherein disrupted attention has proximal effects on interactive behavior that interferes with learning and has distal, cascading effects on long-term outcomes.
3. **Lead to the development of new models for earlier intervention** that enrich early experiences and enhance learning opportunities *from the first months of life*, ultimately reducing lifelong costs and improving lifelong outcomes for individuals with ASD.

There is an urgent need to test specific mechanistic hypotheses about precisely *when* attention patterns diverge in ASD, the *neurobiological systems* implicated in this disruption, and the *in-the-moment* influence of attention on interactive behaviors in infancy, *prior to the emergence* of symptoms.

The First Three Months: A Critical Period of Neurodevelopment and Autonomic Maturation

The brain undergoes faster structural and functional maturation, as well as significant cross-domain integration, from birth to 3 months of age than at any other postnatal time[47]. From birth, the autonomic nervous system (ANS) plays a vital role in the adaptive competence of the newborn. ANS has two major component systems: sympathetic ("fight or flight") and parasympathetic ("rest and digest"). In response to ever-changing environmental contexts, these systems enact physiological changes that regulate basic functions, such as heart rate and respiration. In early prenatal development, physiological functions are driven primarily by sympathetic influence. Higher cortical processes begin to integrate autonomic control in the late prenatal and early postnatal period leading to increasing influence of the parasympathetic system. As ANS matures, parasympathetic tone serves as the foundation for several higher-level behaviors that support calm states of alertness and **sustained attention** to people and objects, ultimately providing countless opportunities for social and language learning. In this way, ANS is considered a "developmental resource" in the neonatal and early infant period that mediates infant-environment interactions[11]. While some neurodevelopmental models of ASD suggest atypical emergence of cortically-driven endogenous attention systems[48], **the developmental course of parasympathetic influence on attention in ASD has never been investigated,** representing a clear scientific gap to be filled[9,48,49,11]. Current evidence that forms the **scientific basis of this study** includes: 1a) *very early* disrupted attention, measured behaviorally, in ASD, 1b) an ANS-regulated marker of attention that is well-studied in TDs, but not ASD, 2) a developmental profile of PT infants who experience very early ANS dysregulation, but not ASD (in most cases), and 3) some evidence of a dynamic, temporally coordinated, ANS-attention-behavioral system in TDs, but less in ASD. This strong scientific foundation (described in further detail below) **calls for further research** on the "physiological phenomena" of attention[49] and presents **<u>three specific gaps</u> in prior research** that prevent a full understanding of the role of autonomic regulation in atypical attention in ASD: **(1)** The maturational course of autonomically regulated attention and baseline autonomic functioning as it emerges *within the first three months of life* for infants at elevated genetic likelihood for ASD remains unknown. **(2)** The developmental course of (and relations between) autonomically regulated attention, baseline autonomic functioning, and ASD symptomology in infants who are *born with* baseline ANS dysfunction (PTs) has not been delineated. **(3)** The dynamic role of HDA in supporting *interactive behavior,* moment-by-moment, is entirely unexplored. These gaps have prevented our ability to test a full model of ASD wherein early autonomic dysregulation of attention *within the first months of life* has cascading effects on interactive behavior and the emergence of ASD symptoms. Examining this novel mechanistic hypothesis while capitalizing on an especially critical period of rapid neurodevelopment (1-3mos) will advance scientific knowledge of ASD pathogenesis, ultimately paving the way for novel paradigms of very early interventions.

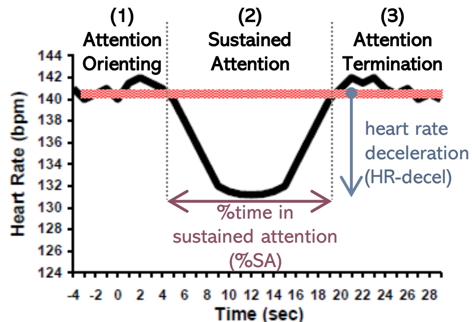


Figure 1. Heart defined attention phases and proposed DVs: %SA, HR-decel

1. Autonomic Regulation of Attention: A Critical Opportunity for ASD Research. Autonomic regulation and attention are critical components of an adaptive engagement system that selects relevant features of the environment and creates rich opportunities for learning and social interaction[10]. ANS serves a regulatory function in attention by modulating heart rate responses during periods of attention and information processing. In a healthy ANS-mediated attention system, a neonate overrides metabolic demands to generate a physiological response (heart rate deceleration) that facilitates and enhances attention and stimulus intake[50]. This parasympathetically-driven physiological response, termed here **heart defined attention** (**HDA**),[51] is characterized by 3 discrete attention phases (*Fig 1*) that are associated with cognitive and learning processes[52]. Some argue that HDA indices are more precise measures of information processing than looking time alone[53–55]. Limited studies of infants later diagnosed with ASD show atypical behavioral measures of attention[7,43,56] in very early infancy (see PI's work, *Fig 2*)[57] as well as atypical general autonomic functioning in later infancy[58,59], pointing to the possibility of autonomically-driven dysregulation of attention. **Our pilot work**, and that of others, show differences in HDA quantity (**%SA:** %time in heart-defined sustained attention) and quality (**HR-decel:** magnitude of heart rate deceleration**;** *Fig 1*) in infant siblings of children with autism (**ASIBs**)[24,49] (see *Fig 4* below).

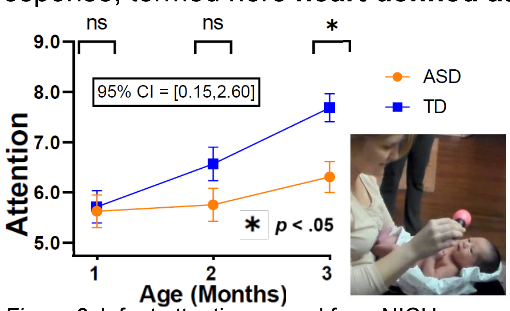


Figure 2. Infant attention scored from NICU Network Neurobehavioral (NNNS) exam. Higher score indiciates better performance.

2. Preterm (PT) Infants as a Model of Global ANS Dysfunction. Significant ANS maturation occurs in the second half of pregnancy, and preterm birth (born <37w gestation) significantly disrupts ANS development and the overall functioning of the PT newborn[60]. For PTs, reduced ANS activity fails to normalize at term age equivalent[25,61] and may alter long-term temporal programming of ANS maturation[62]. While the vast majority of PTs experience prolonged global ANS dysfunction, *only a small minority of PTs develop ASD* (7% in PTs vs. 20% in ASIBs)[16,17]. Thus, inclusion of **PTs as a model for global ANS dysfunction** in this study allows for heightened precision and specificity in delineating the link between *attention-specific* ANS dysfunction (i.e., HDA) and ASD symptoms. **Systematic, longitudinal examination of autonomic measures of attention (HDA) in infants with elevated liability for ASD (ASIBs) contrasted to PT and TD control groups** is necessary to test our mechanistic hypothesis that disrupted autonomic regulation <u>*specific to*</u> attention (vs. global ANS dysfunction observed in PTs and measured in this study) carries critical consequences for learning and development[63] and is associated with ASD symptoms. Measurement of general, baseline autonomic function in addition to attention-specific autonomic regulation, as proposed in this study, will clearly show how autonomic regulation of attention differs across **ASIBs, PTs,** and **TDs**.

3. Modeling Autonomic, Attentional, and Behavioral Dynamics in ASD. A healthy, regulated autonomic and attention system creates rich opportunities for learning and social interaction[10]. After the third month of life, a newly emerging, cortically driven organization of behavior results in anticipation and preparation for interactions with people and objects. The dynamic coordination of autonomic arousal and attention affords calm states of alertness, an optimal context for learning[64]. Quantification of the dynamics of this system[65] can reveal how autonomic regulation of attention is associated with, and may even predict, interactive behaviors as they unfold, moment-by-moment, in infancy. Autonomic arousal can index "anticipatory readiness" for dynamic changes in attention and behavior[66,67]. Emerging research in ASD suggests unique patterns of ongoing physiological activity and attention during social tasks[68,69], however, **there is a significant lack of data on temporal dynamics of HDA and ongoing behaviors**[70] **in ASIBs** in infancy as symptoms of ASD begin to unfold. Such ANS-driven dynamics could facilitate ongoing social engagement, which may help to guide ongoing interactive behavior, and ultimately increase opportunities for learning. **The proposed study will fill this gap** with an analysis of temporal dynamics of infant gaze, heart rate changes, and behavior during naturalistic interactions between age 6-12 months across ASIBs, PTs, and TDs. Critically, the direct link from ***physiology*** (heart-defined attention) to **in-the-moment interactive *behavior*** will enable translation of findings to pilot behavioral interventions and explore biofeedback tools[19,71,72] to directly target the quality and quantity of infant HDA during interactions.

Model of ANS-Regulated Attention and ASD

Based on current literature and our preliminary data, we propose a model of ASD wherein initial global ANS (primarily sympathetic) function supporting neonatal arousal and reactivity is intact at birth. Yet, subsequently emerging parasympathetic regulation of attention is disrupted. This disruption emerges over the first year of life and interferes with infant sustained attention and ensuing behavior, leading to decreased learning, especially social learning, opportunities, and increased ASD symptoms. **Expected Outcomes by Group** (see *Fig 3):* **ASIBs**, who have *no/low* global ANS dysfunction at birth (*Fig 3a*), will show emerging ANS dysregulation *of attention* over the first year (attenuated HDA from 1-3m, weakening HDA-behavior associations from 6-12m; *Fig 3b*) leading to elevated ASD symptoms at age 3 (*Fig 3c*). **PTs**, who *do* have global ANS dysfunction at birth (*Fig 3a*), should show gradual improvement in HDA (1-3m) and HDA-behavior associations (6-12m), ultimately resulting in low ASD symptoms at age 3. The rate of ASD in PTs is higher than that of the general population, and we hypothesize that a small subset of PTs with elevated ASD symptoms at outcome will show *both* disrupted global ANS *and* ANS-dysregulated attention in early infancy. **TDs** (*no* global ANS dysfunction) will show no/low ANS dysregulation of attention and low ASD symptoms at age 3.

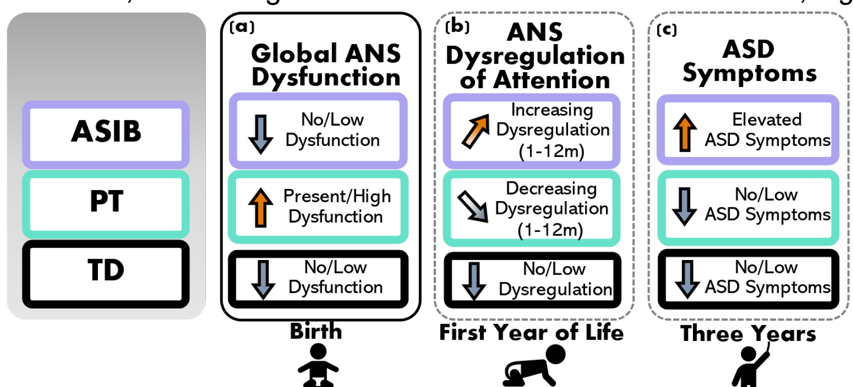
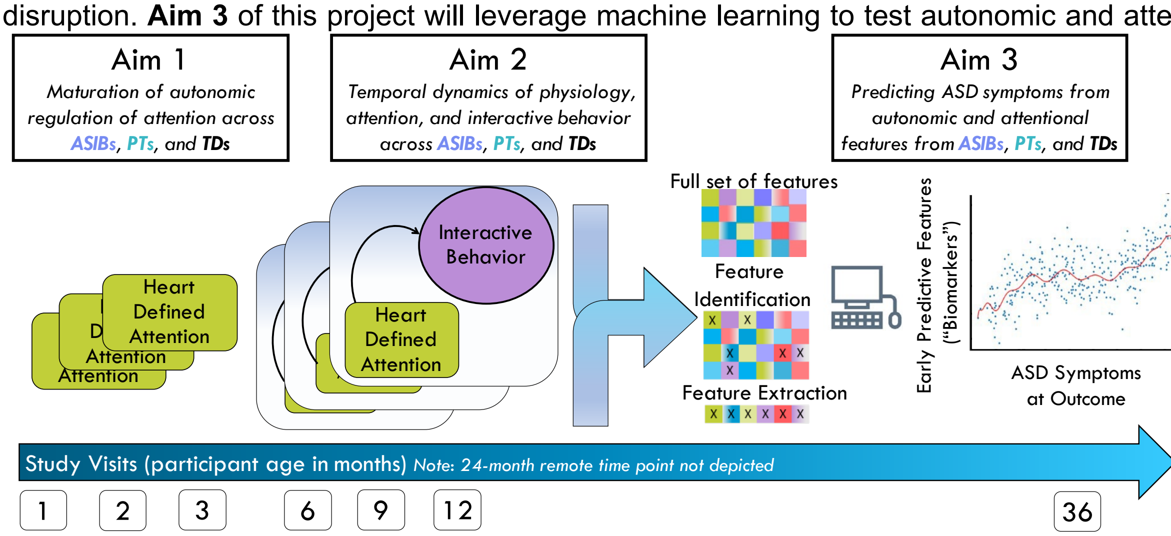


Figure 3. Driving framework for expected outcomes to be tested (b) and (c) given initial global ANS function/dysfunction at birth (a). ANS dysregulation of attention (b) will be measured measured early (1-3 months) via heart defined sustained attention (%time in sustained attention, HR deceleration). ANS dysregulation of attention will be measured later (6-12 months) via HDA-behavior associations.

Translation of Findings to Non-invasive Biomarkers of the ASD Phenotype

Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the child, enhancing quality of life for the family, and reducing lifelong financial burden[73]. Very early onset of intervention enhances child gains, which has prompted the PI and others to develop and pilot interventions for at-risk infants, some *as young as 6 months*[31,74–77]. The nascent, but rapid, progress in the development of very early interventions demands innovative studies to identify early markers of social functioning. Offering intervention to all infants with genetic or congenital risk is likely not feasible. But a **valid, cost-effective, non-invasive biomarker** can help identify those who have the greatest likelihood of developing social-communication challenges and ASD, and thus are in greatest need. In a landmark study, Jones and Klin (2013)[7] identified altered trajectories of social attention (2-6 months) as the earliest known predictor of ASD. However, this exceptional study did not evaluate potential biological mechanisms underlying such an early disruption. **Aim 3** of this project will leverage machine learning to test autonomic and attentional features from 1-12 months (Aims 1-2) as viable biomarkers of ASD symptoms. Notably, **Aim 3 is not contingent** on significant results from Aims 1-2 as statistical significance is not a requisite for including these variables in feature selection



procedures. See **study overview** above.

The proposed research will have a significant impact on the broader understanding of ASD by utilizing an innovative, developmentally informed framework: autonomic regulation of attention[78]. We will generate linkages from ANS and attention to moment-by-moment infant behavior and utilize advanced analytic methods to predict ASD symptoms years later. The prior research and our preliminary data (see *Prelim. Data* below) provide support for the overall study premise to identify disrupted autonomic regulation of attention as an early indicator of ASD.

The potential for translation of findings to actionable interventions is high; promising studies of behavioral interventions for PT infants show positive maturational shifts of autonomic regulation[79], suggesting malleability of this early-developing system. Developmentally informed interventions that target this rapidly maturing autonomic system of attention have the potential to make significant, sustained changes in attention, cognitive, and social outcomes for infants with an elevated likelihood for ASD.

INNOVATION

This proposal seeks to shift current research paradigms by studying novel indices of attention across three unique infant groups using advanced modeling of dynamic, moment-by-moment, behavior in the first year of life.

Attention as an Autonomically-Regulated Process and Predictor of ASD. Atypical attention has been among the most promising early markers of ASD[38,80], but the biological underpinnings of disrupted infant attention in ASD is currently unknown. This proposal will utilize <u>an innovative, physiological measure of attention (HDA)</u> in addition to a baseline autonomic marker (respiratory sinus arrhythmia, RSA) to test the hypothesis that disrupted attention is linked specifically to altered ANS-regulated attention and is associated with later-emerging ASD symptoms. Repeated measurement of RSA and of HDA across interactive contexts from the first months of life is an innovative approach that will provide an enriched dataset more powerful than previous studies. This innovation will allow us to understand *mechanisms* and *roles* of ANS and attention in the unfolding of ASD.

Preterm Infants: Controlling for Global ANS Dysfunction. PT infants are born with an exceptionally immature autonomic regulatory system that may not fully develop until well after birth[13,81–83]. Yet, the vast majority of PTs ultimately experience typical outcomes (~80-92% of PTs do not have ASD or developmental delays by preschool age[84,85]). Thus, <u>PT infants serve as a unique control group that experience early autonomic dysfunction *and* mostly typical, non-ASD outcomes</u>. Measures of HDA and baseline RSA within this innovative three-group research design will allow us to delineate: 1) how global ANS dysfunction (in PTs) differs from autonomic dysfunction specific to attention (in ASIBs) and 2) how maturation and improvements in ANS over the first year of life for PTs may support interactive behavior and learning, potentially buffering against ASD-specific outcomes.

Moment-by-Moment Measures of a Dynamic, Interactive System. Dynamic biological rhythms are central to the development of the infant[9]. Physiological and social maturation are dynamic processes, and variability within this system during critical neurodevelopmental windows may predict later functioning[9,86]. <u>This proposal examines novel questions of how physiological correlates of attention (i.e., HDA) relate to infant interactive behavior.</u> By measuring HDA during live interactions with people and objects, we will explore the dynamics of attention, physiology, and behavior in an integrated way that has never been done before. Aim 2 will capture this integration, generating a rich dataset that can inform the development of interventions.

APPROACH

Key empirical support for the proposed project include the PI and investigative team's preliminary data and previous research on disrupted attention in infants who later develop ASD[6,44,45,48,87–89]. However, prior research has been largely limited to behavioral measurement of attention in contrived experimental contexts (e.g., eye tracking) in later infancy (after 3-6 months). Moreover, very few studies have directly compared ANS or attention in ASIBs and PTs. And no studies have examined the temporal dynamics of autonomic regulation of attention and interactive behavior in these groups. These issues limit advancement of etiological theories of ASD and implications for treatment. This proposal details specific plans to address the gaps in prior research.

Research Team. The project PI (Bradshaw) has >14 years of experience conducting research in largescale, NIH-funded, interdisciplinary, autism centers (Yale Child Study Center, UCSB, UCSD, Emory), resulting in 40 publications. As an early-stage investigator (ESI), the PI is currently **leading** two NIH-funded studies (K23, R21), one Foundation award (James S. McDonnell Foundation), and two internally funded pilot studies. In the proposed project, the PI will be supported by five Co-Is and one psychologist (below), four of whom are senior investigators with an established track record of grants management (R01s) and successful mentorship of and collaboration with ESIs. In addition to regular individual meetings between the PI and Co-Is, **regular meetings with the full investigative team** (PI+Co-Is) will occur at key points throughout the project (≥3x/year) to troubleshoot issues and think through larger project questions and plans from multiple perspectives. **Jane Roberts** (expertise in autonomic dysfunction in ASD/Fragile X; primary mentor for PI's K23) has significant grants management experience and publications with the PI[90]. **John Richards** (co-mentor on PI's K23, collaborator on internal pilot grant) developed the theoretical model of heart-defined attention[52,91] and has led an R01 for >40 years focused on this model of infant attention in TD and PT infants. **Robin Dail** (expert in ANS in PT infants, collaborator on internal pilot grant) is leading a multisite R01 with PT infants in NICUs across North/South Carolina. Her expertise is critical for understanding ANS maturation in the context of PT birth and heightened medical morbidities. The PI's collaborative conceptual work with **Jana Iverson**[8] (expertise in TD and ASIB infant motor/communication development) on developmental cascades in ASD has led to the novel questions proposed in this study. **Christian O'Reilly** (expertise in computational modeling, early biomarkers, and signal processing) has worked with the PI using machine learning to predict ASD likelihood from heart activity (preliminary data for Aim 3). Finally, a licensed **Clinical Psychologist** with expertise in developmental psychology and autism (Dr. Emily Neger) will assist the PI with clinical training, supervision, reliability monitoring, and expert clinical judgements. The unique, complementary areas of expertise among the investigative team, along with diversity across stage of career and discipline, will enhance diverse perspectives, enrich expertise, foster ingenuity, and result in successful achievement of scientific aims.

PRELIMINARY DATA

ASD-Specific Differences in Behavioral Measures of Early Infant Attention. In the neonatal period, attention to the environment occurs during calm states of alertness, facilitated by optimal levels of autonomic regulation. Our recent set of studies (P50MH100029)[6,57] investigated *behavioral* measures of **attention** in 1-3-month-old ASIBs who later developed ASD. We identified specific timepoints, between 2-3 months of age, when maturation of attention in ASD (n=18) and ASIBs (n=41) lagged compared to TDs (n=39; *Fig 2*, above). These preliminary data highlight **feasibility** in **recruitment** and **measurement** of attention in neonates and **quantifiable differences** in behavioral measures of attention across groups. The current proposal will expand on these data by increasing sample size (n=75 ASIBs), adding a second comparison group (n=75 PTs), and examining a specific biological substrate of disrupted attention (heart-defined attention; HDA) as well as general autonomic integrity (baseline RSA). In addition, all data collection will occur in the home, which will greatly increase participant accessibility and enrollment (especially for communities of color and rural populations).

Infant Autonomic Regulation of Attention is Associated with ASD Symptoms. Preliminary data related to Aim 1 include trajectories of HDA during naturalistic parent-infant interactions from 1-4 months of age across n=10 ASIBs, n=20 PTs, and n=30 TDs (K23MH120476; R21DC017252; PI: Bradshaw). These trajectories are characterized by a significant increase in heart rate (HR) deceleration during sustained attention ($F=10.24$, $p<.001$, *Fig 4a*) and point to a dampened HR deceleration that is developmentally transient for PTs, but persistent for ASIBs[23].[949260] Moreover, HR deceleration during sustained attention to people at 3 months is associated with early-emerging symptoms of ASD[26] (social-communication skills at 9mo[92]; $r=0.50$; $p<.05$; *Fig 4b*). Our team's past research also suggests dampened baseline RSA is present very early in PTs[93], but only later in ASD[90] (similar to other work[58]). Building on these preliminary findings, we propose to test the hypothesis that autonomic dysregulation of attention is present early in ASIBs (Aim 1), disrupts temporal dynamics of physiology, attention, and interactive behavior (Aim 2), and is associated with elevated ASD symptoms at age 3 (Aim 3).

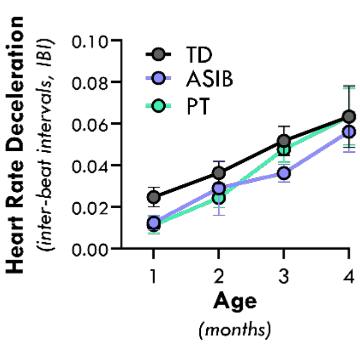


Figure 4a. Trajectories of HDA deceleration during social interactions.

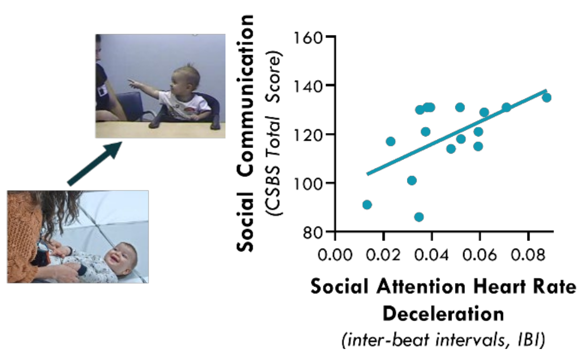


Figure 4b. Association between HDA deceleration during social interaction at 3-4mos and 9mo communication.

Feasibility of Time-Locking HDA Phases to Interactive Behavior. Our data show that HDA-related HR deceleration during a social interaction at 3 months is associated with social-communication outcomes at 9 months[92] (see above, *Fig 4b*). One interpretation of this preliminary finding is that more sustained attention during social interaction reflects increased social engagement and learning, leading to better social-communication skills 6 months later. Yet these data also point to questions about whether and how increased autonomic regulation of attention may facilitate interactive behavior (and therefore learning) ***in the moment***. Aim 2 will address this question. Utilizing full electrocardiogram (ECG) waveform recordings (1024Hz) and frame-by-frame video coding of gaze fixation, ongoing heart-defined phases of attention (HDA) are identified epoched based on the onset of interactive behavior (also coded frame-by-frame from video). Our preliminary data (K23MH120476) demonstrate **feasibility in capturing infant interactive behavior in 6-12-month-olds**[26,94] **and synchronizing with HDA time-series data**[23].

In a preliminary analysis of N=12 infants, we identified the onset of interactive behavior (object exploration) and calculated the proportion of time (i.e., relative frequency) of HDA phase in a time-series of 5 seconds prior to each behavior onset (*Fig 5)*. Findings point to between-group differences (TD vs ASIB) in autonomic regulation of attention immediately preceding behavior. As expected, TD infants showed a consistent pattern of HDA change within the 2 seconds immediately preceding the behavior**,** characterized by an increase in attention (increased **attention orienting**, green and **sustained attention**, red). By contrast, **ASIBs did not show a clear pattern of HDA change** immediately preceding the behavior. Instead, attention patterns remained stable up to 5 seconds prior to the behavior, characterized by less overall attention (more **inattention**, grey; less **attention orienting** and **sustained attention**). The proposed project will build upon this preliminary work in a rigorous examination of **moment-to-moment dynamics** of HDA and interactive behavior across the first year.

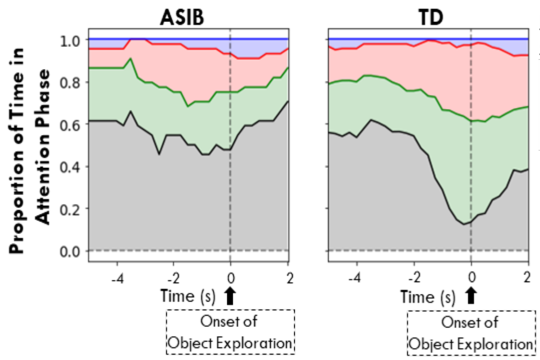


Figure 5. HDA phases (relative frequency of each attention phase) leading up to object exploration behavior for 9-month-olds.

Prediction of ASD Symptoms using Measures of ANS and Attention. In collaboration with Co-I (O'Reilly), ECG data from 3-6-month-old ASIBs (N=7) and TDs (N=16) during interactions were collected and cleaned. Several features were extracted (e.g., mean inter-beat-interval (IBI), standard deviation, root-mean-square differences between adjacent IBI, discrete functional analysis, sample entropy) and an ablation study was used for optimal feature selection. A multilayer perceptron was used to classify ASD likelihood (ASIB vs. TD) with 86% accuracy (precision: 65%, sensitivity: 65%, specificity: 94%). These preliminary results compare advantageously with a previous study reporting a 75.3% accuracy[95] and **demonstrate scientific and technical feasibility** in feature extraction and application of machine learning models to predict ASD likelihood. In this proposal, we will develop a more nuanced analysis predicting ASD symptoms on continuous scales using newly developed deep neural implementation of mixed-effect linear models[96]. The proposed project will use autonomic and attentional features collected for Aims 1 and 2 to better understand how HDA can be leveraged in the formulation of biomarkers in ASD. **Our preliminary data suggest sufficient power** for applying these methods to a sample size that is over ten times larger than that of this pilot study (N=200; n=75 ASIBs). Further, large effect sizes (detectable on relatively small samples) are critical for developing effective ASD biomarkers.

Regression Over Continuous Scales Rather than Binary Diagnoses Classification. ASD symptoms will be captured using Autism Diagnostic Observation Schedule (ADOS-2) calibrated severity scores (CSS)[97]. Our preliminary data (PI Bradshaw[57] and Co-I O'Reilly) suggest sufficient variability in CSS scores across groups of 3-year-olds with and without ASD to determine meaningful associations between early infant biomarkers and ASD symptoms. In the current proposal, we will further build on this direction using ANS-related predictors.

OVERALL EXPERIMENTAL DESIGN

Participants. A total of 200 infants will be included in this study: 75 ASIBs, 75 PTs, and 50 TDs. **Note that our analyses are not powered to detect associations between predictors and binary ASD classification. Rather, for Aim 3, our sample size is powered to detect associations with a continuous measure of ASD *symptoms* at age 3** (see *Power Analysis* below). Our power analyses, published studies,[57] and preliminary data suggest that this proposal is sufficiently powered to detect associations of infant biomarkers and ASD symptoms[6,57,92]. All participants will be followed longitudinally from 1 month to 3 years, at which point **stable** ASD symptoms can be quantified using ADOS CSS[98,99]. In accordance with NIH guidelines for consideration of **relevant biological variables,** we acknowledge that ASD prevalence is higher in males; however, males and females will be recruited equally, and sex included as a covariate in all analyses.

Inclusion Criteria: 1) **ASIB:** born fullterm (≥37w), have a full biological sibling with ASD (confirmation of ASD in the older sibling: record review, direct clinical measures (SCQ[100], CBCL[101], SRS-2[102]), expert clinical judgement by licensed psychologist with expertise in ASD); 2) **TD**: born fullterm, have a full biological TD sibling (confirmed with SCQ, CBCL, SRS, expert clinical judgment), no history of ASD in 1^{st}-3^{rd} degree relatives; 3) **PT**: gestational age (GA) at birth between $24_{0/7}$–$32_{6/7}$ weeks, have a sibling, no family history of ASD in 1^{st}-3^{rd} degree relatives. **GA Considerations:** PTs will be evenly stratified into 3 GA bins to ensure relatively even distribution of risk for ANS dysfunction: (24-26w; 27-29w; 30-32w). GA at birth will be included as a covariate in all analyses, similar to our previous work[57,103]. **Exclusion criteria** for all participants: failed newborn hearing screen, known major hearing or visual impairments, non-febrile seizure disorders, known genetic/chromosomal syndromes (e.g., Down Syndrome), congenital heart condition known to effect heart rate, severe brain hemorrhage and periventricular leukomalacia (PVL). PTs will provide medical records for accurate eligibility assessment. To maximize recruitment efforts of PTs, we are not excluding twins (but multiple birth will be considered in analyses). TDs or PTs with an ASD diagnosis at outcome will be included in analyses (while considering GA and genetic likelihood in models) to increase generalizability.

Feasibility of Recruitment/Enrollment. The current project capitalizes on significant, established recruitment infrastructure, which has led to successful recruitment of ASIBs and PTs at USC. Resources include: the UofSC Neurodevelopment Registry (N>1,100; PIs: Roberts and Bradshaw), the USC Infants and Children Registry (N>150; PI: Richards, Bradshaw), multiple community partnerships (NICUs at the two largest hospital systems in the state, SC SPARK, CAN Center at USC, DDSN, Pediatric practices, autism services providers). USC investigators have **<u>direct outreach access to all children with ASD in the state</u> of South Carolina (total N>8,000**; n=6,700<8yrs) through the Department of Disabilities and Special Needs (DDSN). This outreach has been exceedingly successful for recruitment in the past (see *Letters of Support*). Our longitudinal studies (K23MH120476, R21DC017252) reflect consistent, successful recruitment and retention of ASIBs, PTs, TDs, **1-3 months of age** with **>95% retention**. The PI's lab consistently receives **30-50 referrals *per week*** from interested research participants in the local community and the average enrollment rate for ASIB studies at USC is **12-18 participants/month.** For 1-3-month-old ASIBs specifically, active recruitment in the PI's lab and at USC results in **enrollment of 4-6/month**. This track record demonstrates feasibility regarding enrollment for the proposed project (target average:10-15 enrollments/month across all groups). Our outstanding recruitment records are bolstered by established connections to supportive networks in South Carolina (ASD service providers, schools, pediatricians, hospital NICUs) and state organizations serving ASD and PT populations have pledged tremendous recruitment support (see *Letters of Support*). **To maximize enrollment, we will leverage in-home data collection** for study visits**.** The investigative team has ~20 years of experience with in-home data collection (TDs, PTs, ASIBs age 1wk-3yrs; N>450) and our data evidence that 99% of families (446/450+) prefer in-home visits when given the choice of home vs. lab. This approach will significantly increase participant recruitment/retention rates and **support access for families with limited resources or located in rural areas.**

Attrition. Resources for participant incentives and study visits include over-enrollment to account for expected attrition (rate of 5% based on current longitudinal studies of ASIBs, PTs, and TDs enrolling at 1-4 months). This includes an additional N=16 participants to account for attrition (~5% lost to follow-up, n=10). Thus, power analyses (described below) are based on the final expected sample of N=200.

Operational Considerations. The current project will capitalize on existing infrastructure and resources from the PIs ongoing NIH awards (K23MH120476, R21DC017252), foundation grants (James S. McDonnell Foundation), institutional awards, and institutional funds. The PI's lab is staffed with 1 research assistant professor, 1 postdoc, 4.5 research specialists, 1 psychologist, 1 fulltime recruitment coordinator, and 4 graduate and 20 undergraduate students. Two current RAs are trained in most proposed data collection and processing procedures and will be dedicated to this project. Two more RAs will be hired specifically for this project (see *Budget Justification*). The PI and lab psychologist are licensed clinical psychologists and at least two research staff are trained and reliable in all clinical measures.

EXPERIMENTAL PARADIGMS AND DATA COLLECTION

Data Collection Overview*.* A key component of this longitudinal study is dense sampling of infant behavior to capture change during a rapid period of neurodevelopment. Accordingly, study visits will occur at months 1, 2, 3, 6, 9, and 12, with an additional 24-month questionnaire-only check-in (which will also support retention), and a 36-month evaluation of ASD symptoms and overall development. <u>Adjusted age will be used for all **PT** visits</u>. At visit months 1-3, autonomic regulation of attention will be measured with simultaneous video and heart rate recording during administration of the well-established standardized neonatal measure NNNS-II (NICU Network Neurobehavioral Scales-II; see *Early Infant Attention Task* below)[104]. At visit months 6-12, trained research staff will video record two standardized infant interactions. <u>Research staff will be masked to participant group; parents will be reminded at each visit to refrain from discussing any family or medical history with staff</u>. All research staff will be trained to reliability (>80% agreement) in administration and scoring (where applicable) of the clinical and experimental measures; reliability will be monitored monthly by the project psychologist. A comprehensive evaluation of ASD symptoms and overall development will be administered at the 36-month time point. While <u>study aims are not designed (nor powered) to use ASD diagnosis as a binary outcome,</u> a clinical best estimate (CBE) procedure will be used to assign an overall diagnosis at 36 months in order to comprehensively characterize our sample. **ADOS calibrated severity scores** (RRB, Social Affect, Total) will be used as a metric of ASD symptoms for Aim 3, similar to our previous work[57]. In addition, each study visit will begin with a 5-min baseline ECG recording while the infant is laying or sitting still watching a video of fractals with classical music (to encourage restfulness/ reduce movement). As in previous work[58,59], **baseline autonomic function** will be indexed with respiratory sinus arrhythmia (RSA; a measure of cardiorespiratory control that reflects general autonomic integrity[105]; CardioBatch[106].

	CONSTRUCT	EXPERIMENTAL CONTEXT	MEASURES	TIME POINT (MONTHS) 1	2	3	6	9	12	24	36
Aim 1	Autonomic Regulation of Attention	Baseline Heart Activity (5 min); Attention Items (NNNS) (10-20 min)	RSA, HDA	X	X	X					
Aim 2	Moment-by-Moment Dynamics of Physiology, Attention, and Interactive Behavior	Baseline Heart Activity (5 min); Social/Object Interaction (10min/ea)	RSA, HDA, Interactive Behavior				X	X	X		
Aim 3	ASD Symptom Prediction	ADOS (45-60 minutes)	Autism Symptoms (Calibrated Severity Score)	*Features from Aim 1 data*			*Features from Aim 2 data*				X
Clinical Care	Clinical Care for Referrals	ASQ-3 (10 min)	Developmental Milestones						X	X	X

Figure 6. Data collection timeline and primary dependent variables. Note that some important measures are not depicted, including covariates (e.g., NICU-related morbidities, GA, motor skills).

Early Infant Attention Task from the NNNS-II (Aim 1). Attention will be measured from 1-3m during a standardized, readily translatable measure (NNNS-II) that is grounded in a developmental, neurobehavioral framework and aligns with study aims and hypotheses. The NNNS-II is a 15-20-minute direct evaluation of neurological functioning and behavioral adaptation of at-risk (e.g., medically compromised, preterm) and healthy neonates[107]. The PI has >6 years experience administering the NNNS, is a certified NNNS-II examiner and trainer, and has published on attention using the NNNS in TDs and ASIBs[6,57]. Of interest to the current study are the NNNS-II Attention items, which examine the infant's ability to block out external (and internal) stimuli and attend to visual and auditory stimuli. Six attention items include inanimate (ball and rattle) and animate (face and voice) stimuli that are presented visually with sound (rattle; face/voice), visually without sound (ball; face), and auditory only (rattle/voice out of sight). Administration is done with the examiner holding the infant in their lap in a flat or semi-elevated position. To obtain optimal performance, examiners implement handling strategies to keep the infant in a calm, alert state (handling strategies are recorded). Attention item administration = 3-6min.

Infant Interaction Task (Aim 2). Infant HDA and interactive behavior will be recorded during two semi-structured, table-based infant interactions at 6, 9, and 12 months (see *Fig 7* and our work using these protocols in 6-12-month-olds[26,94]). To capture both social and object interactive behaviors, two interactions will be completed. The **Social** Interaction occurs with an examiner and caregiver (ecologically valid context for infants) and is designed to elicit social interaction behaviors, similar to common social-communication measures for infants (e.g., CSBS[108]). The PI is a close collaborator of the CSBS author (Amy Wetherby)[26,75,109] and our research staff are trained and supervised in the CSBS-BS by Dr. Wetherby's research staff. The **Object** Interaction (no caregivers) is designed to elicit independently initiated object exploration behavior. Our research team has successfully administered this experimental task to quantify object behaviors[94]. Social and Object interactions are equated on duration and number of objects presented to the infant (n=5; *Fig 7*).

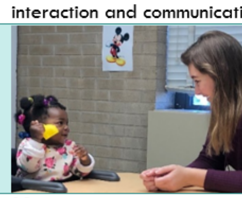
		Social Interaction	Nonsocial Interaction
Protocol Description		Semi-structured interaction between infant, examiner, parent, and one object at a time. Examiner provides presses to elicit infant social interaction and communication.	Semi-structured interaction between infant and one object at a time. Distinct object shapes and functions provide various opportunities to elicit infant exploratory behavior.
Experimental Setup			
Task Length		10 min	10 min
Activities Presented		wind-up toy · squeaky toy · balloon · koosh ball · bubbles	rattle/sensory teether · soft blocks · ring stacker · noise maker · pop-up
Action Examples	Earlier Actions	- Eye contact, Vocalization	- Grasp, Mouth
	Later Actions	- Point, Word Approximations	- Stack, Press

Figure 7. Infant Interaction Protocol

Autism Symptoms (Aim 3). ASD symptoms at 36m will be captured using ADOS-2 calibrated severity scores (CSS): Social Affect, Restricted and Repetitive Behaviors (RRBs), and Total[97]. ADOS examiners will be trained to research reliability and supervised by the PI and project psychologist. When the administering examiner is unintentionally unmasked on a study visit, a masked examiner will score the ADOS from video to ensure scientific rigor. CSSs range from 1-10 and represent an excellent measure of ASD symptoms, relatively independent from IQ and language abilities[97,110].

Data Processing

Electrocardiogram (ECG) Recordings. ECG recordings will be collected at each time point between 1-12 months via Actiheart[111] wireless heart rate monitors, sampled at 1024Hz. The research team has extensive experience using these monitors for successful, continuous recording of infant ECG. Video recordings of all experimental paradigms will be collected for synchronization of ECG recordings with behavioral events (behaviorally coded later from video). Specialized software programs, CardioPeakSegmenter and CardioEditPlus[106,112], will be used for R peak detection, segmentation, and artifact removal.

Baseline Autonomic Function will be indexed using overall RSA from a 5-min baseline recording[106].

Synchronization of ECG and Video Recordings. Actiheart wireless heart rate monitors are incredibly user-friendly and allow unrestricted movement for infants and caregivers. The start time of the ECG recording (based on a remote computer time) is used as the synchronization signal and is captured (video recorded) at the beginning of the experiment (i.e., the start time and run time of the ECG recording is captured on the video that will later be coded for infant behavior). Our custom data processing pipeline then crops the ECG recording to align ECG and Video start times ($time_0$(ECG)= $time_0$(Video)), allowing precise alignment of behavioral coding (described below) and ECG. This method has been used successfully in the PI's lab for the past 4 years[23,92].

Behavioral Coding of Infant Gaze. Coding of infant looking is needed to identify heart defined phases of attention. Thus, frame-by-frame video coding of infant looking using DataVyu software will be completed by trained and reliable undergraduate RAs masked to group. The PI has 16 years of experience in behavioral coding and the PI's lab has established procedures and manuals for coding infant looking during interactions[23,92]. Coders will identify the onset and offset of infant looks to pre-identified target stimuli (object, caregiver, examiner) during NNNS attention items (1-3m) and infant interactions (6-12m). For all interactions, only one object at a time will be presented. Our established training/reliability procedures ensure high research rigor.

Behavioral Coding of Infant Interactive Behavior. The occurrence of infant interactive behaviors will be coded from video by masked RAs (Aim 2). Dr. Iverson (Co-I) is a renowned expert in capturing infant social[113–116] and object[117–119] interactive behavior in PTs[120,121] and ASIBs[114,117] and will assist with this process. Due to rapid development in social and object interaction skills from 6-12 months, behaviors will be categorized as "earlier" and "later" emerging (*Fig 7* above). This will allow our measures to account for maturation in interactive behaviors and avoid floor and ceiling effects at each time point. For example, we expect no infants show "later" social acts at 6m (words and gestures) but many infants show "earlier" social acts at 6m and 12m (social smiles).

HDA. The PI and Co-Is O'Reilly and Richards developed customized Matlab and Python scripts to synchronize heart rate (HR) with infant looking to calculate onsets and offset of HDA phases (sustained attention defined as 5 successive beats with HR slower than pre-look HR median[22,23,92]). Summary variables calculated for analysis will include duration of sustained attention (**%SA**) and heart rate deceleration during sustained attention; these have been associated with enhanced learning[122] and will serve as the primary DVs (Aims 1-2).

Covariates and Demographic Data. Sociodemographic data (race, ethnicity, income, education, home address) and medical history (pregnancy, birth, and early postnatal period) will be collected at the first visit. For PTs, morbidity scores[123] will be derived from medical records and in-depth NICU questionnaire (APGAR scores, overall time in NICU, presence/duration of medical supports (e.g., ventilator), neurological damage (IVH), infections, and surgeries). Due to the potential impact of NICU-related stimulation and stress on cardiorespiratory functions, these variables will be considered as covariates in analyses. Co-I Dail has expertise in evaluating and selecting relevant variables. Annual questionnaires about parental concerns, developmental milestones (ASQ-3[124]), and any early intervention services will be collected. Given the impact of maternal depression on early social development, mothers will complete a depression measure (EPDS[125]) at months 1, 3, and 6. The PI and project psychologist have the clinical expertise to evaluate and address clinical depression and will refer mothers for appropriate care. Motor questionnaires[126] will be completed at each visit to control for impact of motor delays on attention and interactive behavior. Outcome cognitive and motor skills will serve as covariates in all analyses.

Data Management. Management of ECG, video, questionnaires, and behavioral assessments will be led by the Project Manager. Undergraduates will be trained in cleaning and editing of raw ECG files (CardioEdit, CardioBatch), video files will be recorded to SD cards and immediately transferred to our secure server. REDCap will be used for parent questionnaires and to store behavioral data (using double data entry/validation). Original forms will be scanned, stored on a secure server, and shredded to protect participant confidentiality. Experimental data will be processed and linked to demographic/behavioral data for statistical analysis.

SPECIFIC AIM MEASURES AND ANALYSES

Aim 1. Compare the maturation of autonomic regulation of attention in the early infant period (1-3m) across ASIBs, PTs, and TDs.

Overview and Hypotheses. Behavioral measures of attention at 2-3 months reveal significant associations with ASD/ASD symptoms [6,7,57], but biological underpinnings have not been identified. Behavioral measures of looking coordinated with heart rate deceleration reflect focused attention, brain arousal system activation, and enhanced learning[78,122,127]. Thus, HDA in early infancy reflects physiological regulation required to attend to, and learn from, the environment. **Aim 1 will directly test the hypothesis that the maturation of autonomic regulation of attention in young infants from 1-3 months is delayed for PTs and disrupted for ASIBs**.

Measures. Baseline autonomic function will be measured with RSA extracted from a 5-min baseline recording. Autonomic regulation of attention will be measured during the six Attention items of the NNNS-II (see *Data Collection Overview* above). Attention to each stimulus will be coded from video and temporally synchronized with ECG data to calculate: %time in heart-defined sustained attention (**%SA** = time in SA / task duration) and heart rate deceleration in sustained attention (**HR-decel** = maxHR–minHR during each SA phase).

Analytic Plan. We will utilize latent growth curve models (LGCM) to characterize trajectories of HDA (%SA; HRdec) as well as baseline autonomic function (RSA) from 1-3 months. LGCMs can be considered comparable to multilevel models and, in many cases, they are numerically identical, however there are several specific advantages to LGCM approaches[128]. For a single manifest variable measured at a minimum of three time points, an LGCM can be fit to estimate two latent factors, i.e., intercept and slope. This approach is established within a structural equation model (SEM) framework, which allows us to take advantage of the flexibility and broadness of SEM models with the following added benefits: 1) ability to assess global model fit[128,129], 2) easy handling of missing data within and between visits using the full information maximum likelihood strategy[130], 3) inclusion of latent predictor(s)/outcome(s) in model[128,129,131]. Recent research has confirmed that this technique can be applied to small sample sizes with missing data[130]. We will first plot individual trajectories of HDA (%SA, HRdec) to explore differential patterns. The model fit will be determined using chi square tests and fit indices, such as the root mean square error of approximation and the comparative fit index. Mean values of growth factors (intercept, slope) will be estimated and compared across groups (TD, PT, ASIB). We will also examine the variance of the growth factors to test whether individuals share the same starting points and rate of change over time. Covariance between growth factors will provide information about individuals' growth rates related to their starting point. Effects of baseline or outcome characteristics (e.g., gestational age at birth, NICU-related morbidity, biological sex, SES, cognitive outcomes) will be controlled for in all analyses. This is critical as a substantial minority of preterm infants also show cognitive delays[132]. **Power Analysis.** Estimated power for growth curve models was calculated following a simulation-based approach (1000 simulation replicates)[34]. Power rates were computed by assuming 3 measurement occasions and linear growth. Population values used for the simulation were obtained by fitting the linear latent class models (LCM) to our pilot data. Given 200 subjects, power rates to detect individual variability in intercepts and slopes were 0.99 and 0.99, respectively. To detect group differences in trajectories, we computed the power to detect a significant difference in terms of mean slopes between two groups (n=75, n=50) and estimated power was 0.81. Finally, given N=200, we can detect a medium effect (R^2=0.25) when predicting continuous outcomes (ASD severity) with 86% power.

Expected Outcomes and Interpretation. Due to global ANS dysfunction at birth, **PTs** will show reduced baseline RSA and autonomic regulation of attention (less **%SA** and less **HRdec**) from 1-3 months compared to **TDs** ($PT_{1\text{-}3mo} < TD_{1\text{-}3mo}$). Due to *intact* global ANS functioning for **ASIBs** at birth, baseline RSA for **ASIBs** will be comparable to **TDs** from 1-3 months. **ASIB** HDA, on the other hand, will be comparable to **TDs** at 1 month ($ASIB_{1mo} = TD_{1mo}$), but as attention and heart rate become increasingly coordinated over time, **ASIBs** will diverge from **TDs** by 3 months ($ASIB_{3mo} < TD_{3mo}$), similar to our behavioral work[6,57]. Due to the hypothesized attention-specific ANS deficit for **ASIBs**, HDA will be reduced for **ASIBs** by 3 months ($ASIB_{3mo} < PT_{3mo} < TD_{3mo}$).

Aim 2. Determine the moment-to-moment influence of autonomic regulation of attention on interactive behavior across the first year (6m, 9m, 12m).

Overview and Hypotheses. **ASIBs**[26,117,133,134] and **PTs**[135–137] follow unique developmental trajectories of interactive behavior. We have three hypotheses regarding the predictive association between HDA and in-the-moment interactive behaviors: 1) In alignment with our pilot data, **TDs** will show strong, consistent patterns of HDA that predict interactive behaviors, 2) **PTs'** initially immature ANS will cause initially weak predictive associations that improve over time as ANS matures, 3) **ASIBs** will show disrupted associations that weaken (worsen) over time as ASD symptoms emerge. In addition, baseline ANS function trajectories will be compared.

Measures. Interactive behavior will be recorded at 6, 9, and 12 months using a standardized protocol (see above, *Infant Interaction Task*, *Fig 7*). The onset of each interactive behavior will be recorded and synchronized with time series HDA to examine, with fine temporal resolution, the influence of HDA on interactive behavior. Because motor

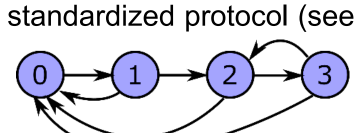


Figure 8. Possible transitions between HDA phases.

differences/delays may influence the frequency/function of interactive behavior[138], visit-specific motor questionnaires[126] will be included as a covariate.

Analytic Plan. We will use HDA times series locked to behavior onsets to study the dynamics of the autonomic attentional system leading up to interactive behavior. We will compare modeling capability of the following alternative approaches: **(1) Continuous-time Markov chain (CTMC):** Our previous work has employed Markov chains to understand dynamics of social attention in toddlers with ASD[139]. A CTMC describes a continuous stochastic process in which the process changes state according to an exponential random variable, moving to a different state with probabilities specified for a Markovian process. We will use this approach to model the process of changing from one HDA phase to the next when moving backward in time, starting from a given action. The structure of the model is fixed given four HDA phases (not looking, orienting, sustained attention, termination) and includes one time constant per state and seven transition probabilities (see *Fig 8*), for a total of 11 parameters. These parameters will be estimated per subject and the distribution of these parameters will be compared between groups. Subject-specific parameters will be estimated in a Bayesian framework using the Markov Chain Monte Carlo (MCMC) approach. **(2) Simple historical mean:** We will compute the inter-beat-interval (IBI) historical mean and corresponding confidence intervals leading up to a behavior. These trajectories will be used to assess group differences in the historical mean and in its time derivatives (speed, acceleration, jerk) to assess group differences in autonomic-attention-action system dynamics. Cluster-based permutation tests will be used to determine periods of time showing statistical differences between groups while accounting for multiple comparisons. This approach is considered as a simpler approach to offer a benchmark for the other, more complex, approach, and previous studies show that simpler approaches often yield better outcomes[140]. **Model Comparison.** We will evaluate the effect size obtained with these approaches to determine which one best captures the dynamics of HDA leading to marked behaviors and extract relevant features (e.g., fitted modeling parameters, historical means over temporal windows, rate of decrease in predictability) as potential biomarkers for ASD symptoms (see Aim 3). **Power Analysis.** Between-group comparisons for the different parameters estimated (i.e., CTMC parameters, IBI means/derivatives over time windows) will be entered into growth models to evaluate this longitudinal data. We therefore expect the same statistical power as Aim 1: range of .81-.99 to detect a medium effect for N=200.

Expected Outcomes and Interpretation: We hypothesize that: (1) baseline autonomic function (RSA) from 6-9 months will be reduced in **PTs** compared to **ASIBs** and **TDs,** but show gradual improvement over time**;** (2) **ASIB** RSA will be largely similar to **TDs** (similar to previous work showing later-emerging RSA disruption in ASD[58]). In terms of HDA-behavior coordination, **PTs** (global ANS dysfunction at birth) will show initial reduced HDA-behavior coordination, and therefore weak HDA-behavior associations, compared to **TDs**. But the effect size of this group difference will decrease over time ($PT_{6mo}<TD_{6mo} \rightarrow PT_{12mo}\sim=TD_{12mo}$). **ASIBs** (hypothesized developmental cascade wherein attention-specific ANS deficits are present early, by 3 months, and contribute to disrupted emergence of interactive behavior[8]) will show increasingly weak associations of HDA with interactive behaviors compared to **TDs** ($ASIB_{6mo}<TD_{6mo} \rightarrow ASIB_{12mo}<<<TD_{12mo}$). **ASIBs'** autonomic-attention-behavior system dynamics may be initially comparable to **PTs** ($ASIB_{6mo} = PT_{6mo} < TD_{6mo}$) due to prolonged ANS dysfunction for **PTs**. But by 12 months, ASIBs will significantly diverge ($ASIB_{12mo} < PT_{12mo} \sim= TD_{12mo}$).

Aim 3. Predict ASD symptoms at age 3 years from infant autonomic and attentional features.

Overview and Hypotheses. Univariate statistical methods for biomarker discovery may be inadequate for capturing the complexity of biological processes involved in a multifaceted, heterogeneous neurodevelopmental disorder[27]. Machine learning (ML) provides unique tools for examining relationships binding multiple interacting variables[28]. Many biomarker studies of ASD (blood, metabolic) have implications for elucidating pathophysiology and improving diagnostic accuracy[141–144], but the lack of conceptual or functional links from **biomarker to behavior** prevents direct translation to behavioral interventions. This aim will incorporate biobehavioral measures of baseline autonomic function (RSA) and autonomic regulation of attention (HDA) to predict ASD symptoms. We hypothesize features collected from 1-12 months will predict ASD symptoms at age 3 years.

Measures. We will use a ML approach considering as features the variables and parameters estimated for Aims 1-2 (regardless of whether the impact of the variables has reach statistical significance for the specific analyses proposed in those aims), as well as simple additional statistics (e.g., overall time in HDA phases) and features established in our preliminary studies (i.e., statistics of heart rate variability, discrete functional analysis, sample entropy from pre-processed ECG with empirical mode decomposition and discrete wavelet transform). As in our previous work[57], ASD symptoms (36m) will be measured using the ADOS-2 Calibrated Severity Scores.

Analytic Plan. To optimize the use of our sample while precluding overfitting and ensuring reproducibility on new data, performance regarding ASD symptom prediction will be cross-validated using a leave-one-subject-out approach. These cross-validated predictions will be compared against the traditional regression approach (i.e., with no test-train split) to assess the degree of overfitting that would result from not splitting the dataset. If overfitting is negligible, this model will be considered final. Else, we will identify the subjects for which the cross-validation prediction is significantly different from the non-cross-validated model, reject the subjects, and re-test. Previous studies using ML in ASD with children and infants concentrated on diagnosis classification. They used various classification methods (support vector machine, k-nearest neighbors, naive Bayes, random forest) and obtained accuracy roughly in the 0.75-0.85 range[145–147]. These studies vary widely with respect to the properties of their sample (e.g., large differences in sample size), their constructs (analysis of electroencephalogram versus spoken language), their objectives (diagnosis in infants versus in children), and their methods in general. As preliminary results, we showed that ANS properties assessed through ECG can be used for prediction by obtaining an 86% accuracy for the classification of ASD risk (ASIB vs TD) using a multilayer perceptron. In this proposal, we will use similar **deep learning models in the context of regression[96] to predict continuous phenotypic outcomes** (autism symptomology; ADOS CSS scores). Continuous outcomes will be assessed through the goodness of fit of regression models (e.g., coefficient of determination, R^2). **Power Analysis.** Sufficient power in our sample is supported by our preliminary demonstration of high predictivity of ECG-derived features for capturing ASD likelihood (classification accuracy=.86 with only N=23). Clinical significance, rather than mere statistical significance, requires large effect sizes (e.g., large R^2). As opposed to p-values, effect sizes do not depend on power and sample size. Reliability of R^2 estimates will be determined using confidence intervals computed with bootstrapping. For a sample size of 200 infants, a single predictor (derived from the combination of many features), and an R^2=0.5, the expected 95% confidence interval (CI) goes from 0.40 to 0.60 (for R^2=0.6, 95% CI = .51-.68). Thus, this study is well-sampled for reasonably narrow CIs on at least medium effect sizes.

Aim 3 Expected Outcomes and Interpretation: We hypothesize that early autonomic-attentional features will predict ASD symptoms that do not emerge until age 2-3 years. Further, through ablation study, we will examine which features from which time points (e.g., 1-3mos vs. 6-12mos) contribute most to predictability. We expect predictability to be highest for attention-specific features and to increase with age.

RIGOR AND REPRODUCIBILITY

Rigor and reproducibility of this project is apparent in our prospective longitudinal design with masked examiners, monitoring of inter-rater reliability/agreement (>80%) for experimental and clinical measures, consideration of potential confounds (GA, medical morbidities, sex), sample size selected based on a priori power analyses, and an investigative team with expertise in grant management/physiology (Roberts, Richards), ANS/prematurity (Dail), infant behavior (Iverson), and signal processing, machine learning, computational neuroscience (O'Reilly).

Potential Problems, Alternative Strategies, & Benchmarks for Success. This project will be initiated nearly upon funding as we have existing recruitment infrastructure and a team trained in most experimental and clinical measures. Activities will occur as follows: recruitment Y1-2, data collection/processing Y1-5, data analyses Y2-5, publications Y3-5 (see *Human Subjects-Study Timeline*). **Recruitment and attrition** are potential concerns for any longitudinal study. Considering past success of the PI and Co-Is in neonatal PT and ASIB recruitment, a plan to see participants in their home statewide, a large research staff, and significant resources dedicated to recruitment, we do not anticipate any difficulty with meeting enrollment targets. We will access institutional funds to support recruitment/retention, dedicate a fulltime recruitment coordinator solely to this project, utilize multiple state-specific autism databases, and employ retention strategies (frequent contact with families through social media and clinical feedback). It is highly possible that **examiners may be unmasked to participant group** during home visits. We mitigate this by: attempting to schedule visits when siblings are out of the home, reminding families to keep examiners masked to family history, collecting data on when examiners are unmasked (to explore in analytic models), process data using masked coders/psychologists (ECG data, attention data, behavioral data, ADOS scores). Our statistical design is **robust to within and between-visit missing data**, we will use state-of-the-art ECG artifact corrections using wavelet and empirical mode decomposition, and we rely on refined longitudinal modeling (i.e., growth curves). In order to ensure robust findings, we will cross-validate our results by employing additional analytic strategies (e.g., mixed effects regression, general linear models) when appropriate. **GA in PTs:** While recent evidence suggests that ANS dysfunction relates more to morbidity than GA[123], we will stratify PT recruitment by three evenly divided birth GA bins (all <32w) to ensure sufficient ANS immaturity. **Significant phenotypic heterogeneity** is expected in ASIB and PT groups. Therefore, comprehensive characterization of ASD and developmental profiles at 36m for all participants will allow for analyses to extend beyond between-group comparisons (Aims 1-2) to include key covariates or stratifiers that can parse out heterogeneity and reveal complex, meaningful interactions. **Aim 3 is not dependent on Aims 1-2** as significant results are not a requisite for including these variables in feature selection procedures (i.e., failure for a test to reach statistical significance does not prove the absence of useful predictive information) and, in fact, may elucidate novel findings above and beyond Aims 1-2.